

University of Windsor

Odette School of Business

The Impact of Extreme Weather on Electricity Demand in Ontario

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1. Introduction and Context

1.1 Background

Extreme weather events occur more often in Ontario. Extreme heat and cold might change consumers' electricity demand, and it will become more difficult for electricity suppliers to identify potential high-demand periods. This change might have a significant impact on electricity supply capacity, resource planning, and grid stability. Therefore, it is essential to help our main stakeholders, including electricity suppliers, grid operators, policymakers, and the government, identify high-demand periods and gain insight into the weather-electricity demand relationship. This project will not only analyze the historical impact of different weather conditions and temperature changes on electricity demand but also develop predictive models identifying unusually high-demand periods and an interactive dashboard displaying the weather-electricity demand relationship in Ontario.

1.2 Project Scope and Boundaries

In scope:

- To select six representative locations across Ontario to analyze and build models to predict other areas in Ontario.
- To collect historical data on electricity demand from IESO and historical data on weather conditions from Environment and Climate Change Canada (2019-2025).
- To perform data cleaning, data integration, data transformation, and data validation using R and RStudio, and conduct in-depth data analysis and predictive modelling.
- To store the cleaned and integrated datasets in MySQL.
- To use temperature as the primary research variable in the data analysis and analyze its potential relationship with electricity demand.
- To evaluate whether the electricity demand-weather relationship will be influenced by other available variables, such as season, date, and region.
- To establish an interactive dashboard through Tableau based on data analysis.

Out of scope:

- To analyze other markets outside Ontario
- To develop a real-time grid management and monitoring system
- To advise electricity suppliers on how much electricity output they need to increase or decrease under extreme weather conditions.
- To consider every factor (such as price, population growth, and changes in consumption habits) that might affect electricity demand.
- To provide electricity infrastructure investment decisions.

1.3 Links to Original User Stories

User story 1: Electricity suppliers and grid operators need to understand the impact of extreme weather on electricity demand in Ontario and identify weather conditions associated with unusual periods of high electricity demand to prepare a resource allocation plan. Their needs will be addressed through data integration, extreme weather analysis, predictive modelling, and data visualization.

User story 2: The project team needs to use credible, clean, integrated, and standardized data and ensure predictive models and the interactive dashboard clearly display analysis results. These needs can be addressed through the functional requirements: data collection, preparation, integration, transformation, and storage, and the non-functional requirements: data integrity, reliability, security, and maintainability.

User stories 3 & 4: Government and policymakers need to understand the weather-related electricity demand patterns to support long-term climate change adaptation strategies and resource planning. Consumers want to gain insight into the impact of extreme weather on electricity demand to enhance awareness of energy conservation and promote sustainable lifestyles. The project will develop an interactive dashboard displaying the weather-electricity demand relationship, extreme weather periods, and predictive model results.

2. Goals and non-goals

2.1 Project Goals

No.	Goals	Measurable success criteria
1	To complete data collection from different sources	We need to successfully collect and import weather datasets from Environment and Climate Change Canada and electricity demand datasets from IESO (Ontario only).
2	To complete data cleaning, data integration, and data transformation	We need to successfully use R and RStudio to ensure data completeness, handle missing values, and make sure key data (electricity demand, temperature, date) have a consistent format.
3	To store data in the relationship database	We need to ensure that cleaned and integrated datasets are successfully stored in MySQL, and sample records can be retrieved through database queries.
4	To complete extreme weather analysis	We need to successfully identify extreme weather conditions and classify extreme and normal weather observations by using predefined temperature thresholds
5	To develop effective and appropriate predictive models	We need to develop predictive models through R and RStudio that can clearly predict and identify unusually high electricity demand periods against a calendar-based baseline.
6	To complete relationship analysis	We need to successfully estimate the weather-electricity demand relationship and predict peak demand using linear regression and quadratic regression, and understand whether other variables (season, date, region) are associated with the impact of weather conditions on electricity demand. All statistical results should be documented and interpreted.
7	To develop an interactive dashboard	We need to develop a Tableau dashboard for data visualization and interaction. The dashboard can display electricity demand and temperature trends using appropriate charts, accurate dates, clearly labelled axes, and measurement units. End users can explore the topic by selecting different criteria, such as time period, season, weather station, and weather condition, in this dashboard.

2.2 non-goals

- To analyze the impact of extreme weather on electricity demand in other provinces.
- To control electricity supply and grid operation directly.
- To develop a real-time grid monitoring and power dispatch system
- To determine whether price, economic development, and population growth are associated with changes in electricity demand.
- To guarantee that predictions will be completely accurate under all weather conditions.

3. System Architecture (High-Level Diagram)



4. Logical Architecture

4.1 Overview

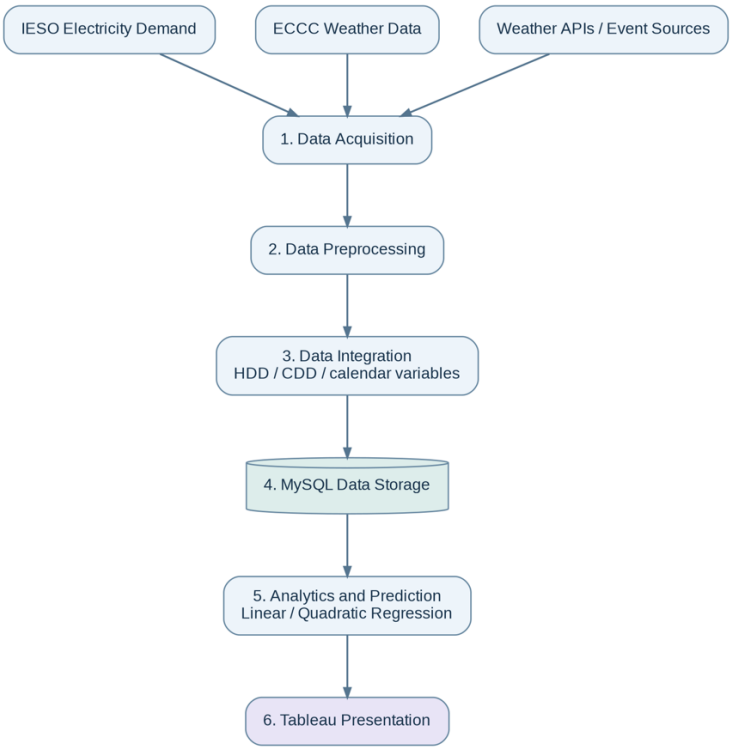
The Ontario Extreme Weather and Electricity Demand Prediction System uses a six-layer architecture to collect, organize, and analyze historical electricity demand and weather data from 2019 to 2025. Each layer handles a specific part of the process, making the system easier to manage and allowing the team to update individual components when needed.

4.2 Logical Architecture Components

Layer	Description
1. Data Acquisition	Collects electricity demand records from IESO, weather observations from ECCC, additional API data, and extreme-weather information from public websites.
2. Data Preprocessing	Uses R to clean datasets, handle missing values, remove duplicates, and standardize dates and measurements.
3. Data Integration	Combines daily electricity demand and weather observations using dates, while calculating relevant weather and seasonal variables.
4. Data Storage	Uses MySQL to organize the cleaned datasets, weather events, and analytical results.
5. Analytics and Prediction	Applies regression models to examine how temperature affects electricity demand and predict unusually high-demand periods.
6. Presentation	Uses Tableau to display electricity demand trends, weather comparisons, and prediction results.

4.3 Component Interaction and Data Flow

Data collected from IESO, ECCC, and other public sources is cleaned and prepared using R. The processed datasets are then combined and stored in MySQL. Regression models use this information to estimate electricity demand, and the results are presented through an interactive Tableau dashboard.



5. Physical Architecture

5.1 Overview

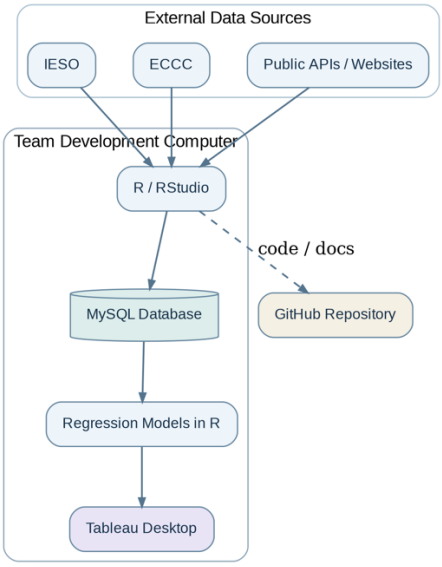
The system is designed to operate on the project team's personal computers without requiring dedicated servers. RStudio handles data processing and modelling, MySQL manages the database, and Tableau is used to present the findings. The system focuses on historical analysis rather than real-time electricity monitoring.

5.2 Hardware and Software Requirements

Component	Technology	Purpose
Development Hardware	Personal computers	Run analytical tools
Programming	R / RStudio	Data cleaning and statistical modelling
Data Processing	dplyr, tidyr, lubridate	Data transformation and feature creation
Data Acquisition	readr, jsonlite, rvest	File imports, JSON processing, web scraping
Database	MySQL / Workbench	Data storage and management
Visualization	Tableau	Interactive dashboards
Version Control	GitHub	Code sharing and documentation

5.3 Physical Deployment Structure

The project follows a straightforward setup. Historical data is obtained from external sources and processed locally through RStudio. The cleaned records are stored in MySQL and used for statistical analysis. Tableau then presents the results through an interactive dashboard. GitHub helps team members manage scripts and documentation throughout development.



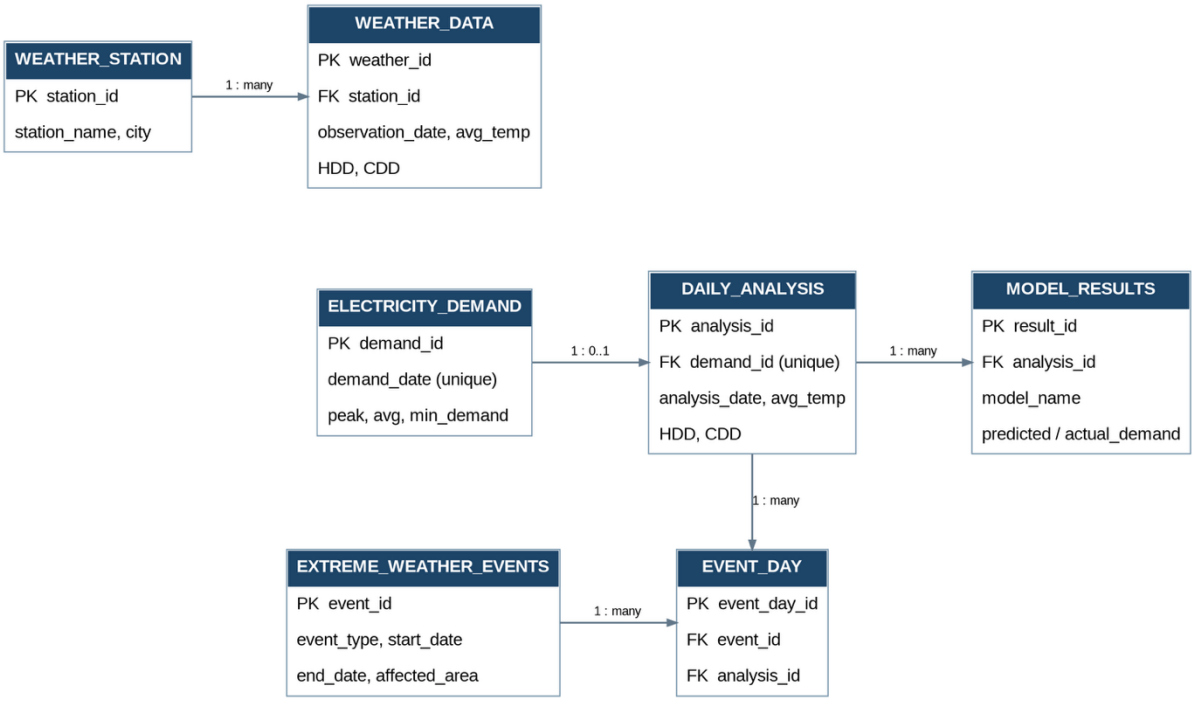
6. Data Model

The project uses a proposed MySQL relational database to organize electricity demand, weather observations, extreme-weather events, and model results. Primary keys and foreign keys help maintain relationships and prevent data inconsistencies.

6.1 Main Database Entities

Entity	Main Attributes
Electricity_Demand	demand_id (PK), demand_date, peak_demand, avg_demand, min_demand
Weather_Station	station_id (PK), station_name, city
Weather_Data	weather_id (PK), station_id (FK), observation_date, avg_temp, HDD, CDD
Extreme_Weather_Events	event_id (PK), event_type, start_date, end_date, affected_area
Daily_Analysis	analysis_id (PK), demand_id (FK), analysis_date, avg_temp, HDD, CDD
Model_Results	result_id (PK), analysis_id (FK), model_name, predicted_demand, actual_demand
Event_Day	event_day_id (PK), event_id (FK), analysis_id (FK)

6.2 Entity Relationship Diagram



6.3 Database Relationships and Integrity

One weather station can provide multiple weather observations, while each daily electricity demand record can be linked to one analytical record. The analytical dataset can also contain predictions from different models.

Extreme-weather events are connected to their relevant analytical dates through the proposed Event_Day table. Weather observations are first aggregated by date before being combined with province-wide electricity demand.

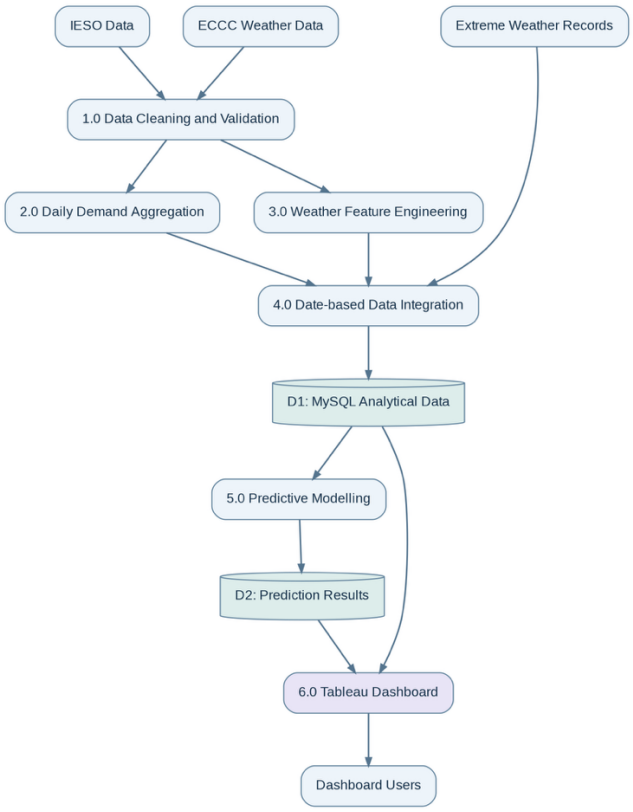
Unique identifiers, foreign keys, and validation checks help maintain the accuracy and consistency of stored information.

6.4 Analytical Data Preparation

R is used to calculate daily peak, average, and minimum electricity demand alongside heating degree days (HDD), cooling degree days (CDD), and seasonal indicators. Daily records with at least four usable weather stations are retained for analysis.

6.5 Data Flow Diagram

The data flow diagram illustrates how information moves from the original electricity and weather sources through data cleaning, integration, database storage, predictive modelling, and visualization.



7. Component Design

The system uses five main processing modules to turn historical data into useful analytical findings.

Module	Function
Data Acquisition	Collect historical demand, weather, API, and extreme-weather event information
Data Cleaning	Identify missing values, remove duplicates, and standardize records
Feature Engineering and Integration	Calculate HDD, CDD, daily demand, and combine datasets
Predictive Modelling	Develop regression models and evaluate prediction accuracy
Visualization	Display historical trends, weather relationships, and model outputs in Tableau

7.1 Prediction Algorithms

- **Linear Regression (Baseline Model):** Provides an initial prediction of daily electricity demand using calendar-related variables such as months, seasons, and weekdays.

- **Quadratic Regression (Proposed Primary Model):** Examines how electricity demand changes with temperature, including the possibility of increased consumption during extremely hot and cold weather.

Model performance is compared using three measures:

- **MAE:** Measures the average difference between actual and predicted demand.
- **RMSE:** Measures prediction errors while giving more importance to larger differences.
- **R^2:** Shows how much variation in electricity demand is explained by the model.

The project may also explore Logistic Regression or Random Forest if additional high-demand classification is required.

7.2 Algorithmic Pseudocode

Two main algorithms describe how the system prepares historical data and generates electricity demand predictions.

Algorithm 1: Data Cleaning and Integration

```
INPUT: ElectricityData, WeatherData
OUTPUT: IntegratedDataset

BEGIN
  REMOVE duplicate records
  STANDARDIZE dates and measurements

  FOR EACH date IN ElectricityData DO
    records ← ValidDemandRecords(date)

    IF COUNT(records) > 0 THEN
      peak ← MAX(records.demand)
      average ← MEAN(records.demand)
      minimum ← MIN(records.demand)

      SAVE DailyDemand(date, peak, average, minimum)
    END IF
  END FOR

  FOR EACH weatherRecord IN WeatherData DO
    IF ValidTemperature(weatherRecord) THEN
      temp ← weatherRecord.avgTemp
      HDD ← MAX(0, 18 - temp)
      CDD ← MAX(0, temp - 18)

      SAVE CleanWeather(weatherRecord, HDD, CDD)
    END IF
  END FOR
```

```
  FOR EACH date IN DailyDemand DO
    weather ← GetWeatherByDate(date)

    IF COUNT(weather) >= 4 THEN
      ADD IntegratedRecord(
        date,
        DailyDemand[date],
        MEAN(weather.avgTemp),
        MEAN(weather.HDD),
        MEAN(weather.CDD)
      )
    END IF
  END FOR

  RETURN IntegratedDataset
END
```

Algorithm 2: Electricity Demand Prediction

```
INPUT: IntegratedDataset
OUTPUT: Predictions, EvaluationResults

BEGIN
  SORT IntegratedDataset BY date

  SPLIT data chronologically
    INTO TrainingData AND TestingData

  baseline ← TrainLinearRegression(
    TrainingData.calendar,
    TrainingData.peakDemand
  )

  weatherModel ← TrainQuadraticRegression(
    TrainingData.calendar,
    TrainingData.temperature,
    TrainingData.temperatureSquared,
    TrainingData.peakDemand
  )

  FOR EACH model IN [baseline, weatherModel] DO
    predicted ← Predict(model, TestingData)

    MAE ← CalculateMAE(predicted, TestingData.peakDemand)
    RMSE ← CalculateRMSE(predicted, TestingData.peakDemand)
    R2 ← CalculateR2(predicted, TestingData.peakDemand)

    SAVE Predictions(model, predicted)
```

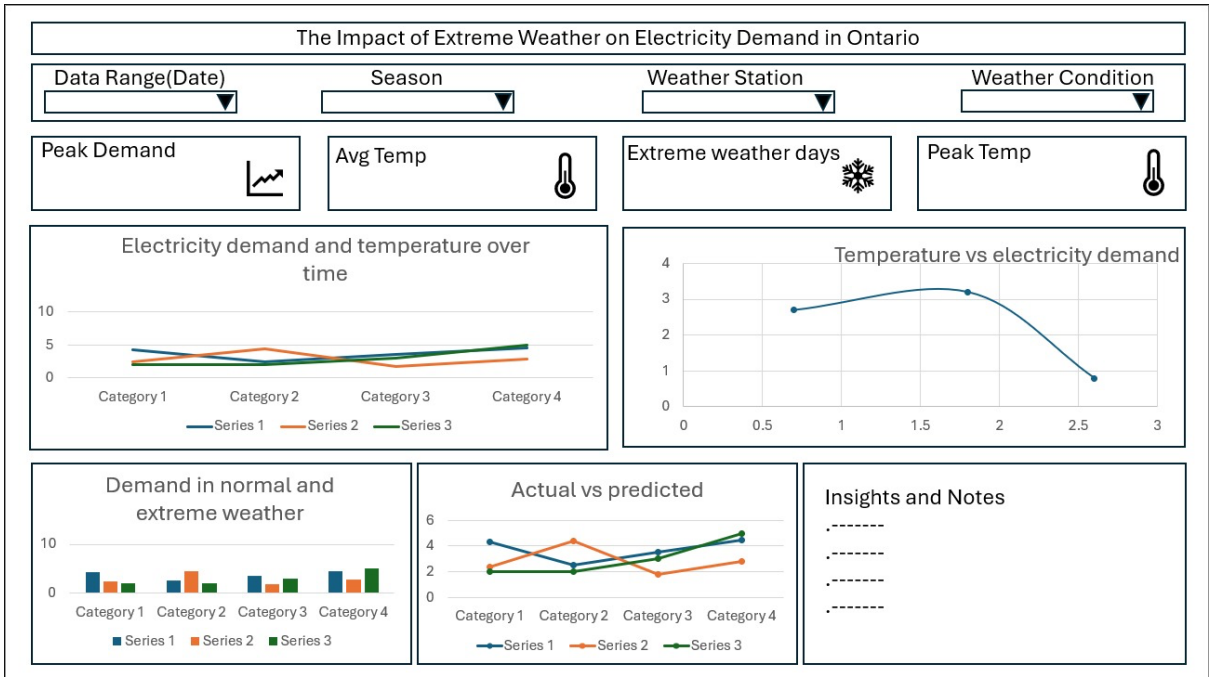
```
    SAVE EvaluationResults(model, MAE, RMSE, R2)
  END FOR

  RETURN Predictions, EvaluationResults
END
```

The algorithms describe the proposed approach to data preparation and prediction. The final models will be evaluated using historical observations before their results are included in Tableau.

8. Interface Design

In this project, we will be using an interactive Tableau dashboard designed for electricity suppliers, grid engineers, and other stakeholders who want to understand the relationship between electricity demand and extreme weather in Ontario. This dashboard will provide date, weather station, season and other filters. For charts and other visualization elements, it will provide electricity demand trends, temperature trends, extreme weather times and model prediction results. All charts will be labelled with measurement units and can be filtered. Users can have a good understanding of the information no matter their technical background.



8.1 Dashboard Interface Layout

Dashboard area	Content
Filters/Header	Shows data range in dates, season, weather station, and weather condition. Users can select their desired range for this dashboard.
Summary section	Shows key data summary such as peak electricity demand, average temperature, extreme weather days, and peak temperature. This section will give users a more direct insight into this analysis.

Trend visualization	Shows the electricity demand and temperature over time. Users will understand that if the temperature is affecting demand in this chart.
Relationship visualization	Shows the scatter plot of temperature and electricity demand. Users will understand how accurate the predictions/ relationship are.
Weather comparison	This section compares the demand for normal, extreme hot and extreme cold weather.
Prediction section	Compares the actual and the predicted data. Users will understand the model's performance results.
Insights and notes	User acknowledgment of what to expect and analysts' notes.

The dashboard will use prepared data from a MySQL database. Then use RStudio to process and build the model for prediction from the original data source and database. After the weather file and electricity -demand files are cleaned and transformed in R it will be stored in MySQL again. Then, Tableau will visualize the data.

8.2 Constraints & Security

The data transformation, processing, and model building will mainly be performed on team members' personal computers. This means the hardware limits will be memory, processing capacity, and storage compared with a professional environment. Consider that the project is using all weather and electricity demand data from 2019 to 2025 for the whole province, the joining, processing and transformation might create performance bottlenecks.

The second constraint is the geographical aspect of the datasets. The different locations selected for this project are all in different areas of the province. These locations may not fully represent all weather situations and conditions in all of Ontario.

The project is already using public data from IESO and the Government of Canada. Therefore, the privacy risk is small. The team will still have basic security measures such as database passwords and API keys, and the team will make sure the data is not stored in the GitHub public repository. Access to the database will be limited to only team members and instructors. Original source files will be saved separately from the cleaned data to ensure the accuracy of the data.

9. Design Rationale

R/RStudio were selected as the main data-processing and modelling tool because the team have experience with R during data analytics class, and it provides strong support for data cleaning, statistical analysis and predictive modelling. With the R package extensions, the team can process data more efficiently.

MySQL was selected because of the multiple types of data used in this project, such as electricity demand, weather observations, locations and analytical data. The relational database function of MySQL allows the team to organize the data with a better structure. It also makes the following visualization process more efficient.

Tableau is selected because of the need for an interactive visual analytics tool rather than text-based and static image output. It supports filtering, charts, and dashboards, which are all interactive for all levels of users. The stakeholders do not need to understand R and SQL to explore the analysis.

Git Hub is an essential tool for code organization, update tracking and team collaboration support.

For the predictive model, the team might use the following:

Linear regression as a baseline model to predict daily peak electricity demand using dates.

Quadratic regression for the main proposed regression model. It predicts demand and captures the nonlinear relationship between temperature and electricity consumption.

Logistic regression for the proposed classification model. It determines whether a day will have high electricity demand with a yes or no answer.

Random Forest for the proposed alternative. It compares classification performance against logistic regression.

10. References

Environment and Climate Change Canada. (n.d.). *Historical climate data. Government of Canada.* https://climate.weather.gc.ca/

Independent Electricity System Operator. (n.d.). *Hourly demand report.* https://reports-public.ieso.ca/public/Demand/